21. Give the symbolic representation of alpha decay, beta decay and gamma emission.
22. In alpha decay, why the unstable nucleus emits ${}_2^4He$ nucleus? Why it does not emit four separate nucleons?
23. What is mean life of a radia active nucleus? Give the expression.
24. What is half-life of a radia active nucleus? Give the expression.
25. What is meant by activity or decay rate? Give its unit.
26. Define curie.
27. What are the constituent particles of neutron and proton?

III Long answer questions

1. Explain the J.J. Thomson experiment to determine the specific charge of electron.
2. Discuss the Millikan's oil drop experiment to determine the charge of an electron.
3. Derive the energy expression for an eletron is the hydrogen atom using Bohr atom model.
4. Discuss the spectral series of hydrogen atom.
5. Explain the variation of average binding energy with the mass number using graph and discuss about its features.
6. Explain in detail the nuclear force.
7. Discuss the alpha decay process with example.
8. Discuss the beta decay process with examples.
9. Discuss the gamma emission process with example.
10. Obtain the law of radioactivity.
11. Discuss the properties of neutrino and its role in beta decay.
12. Explain the idea of carbon dating.
13. Discuss the process of nuclear fission and its properties.
14. Discuss the process of nuclear fusion and how energy is generated in stars?
15. Describe the working of nuclear reactor with a block diagram.
16. Explain in detail the four fundamental forces in nature.
17. Briefly explain the elementary particles present in nature.

Exercises

1. Consider two hydrogen atoms H_A and H_B in ground state. Assume that hydrogen atom H_A is at rest and hydrogen atom H_B is moving with a speed and make head-on collision with the stationary hydrogen atom H_A. After the collision, both of them move together. What is minimum value of the kinetic energy of the moving hydrogen atom H_B, such that any one of the hydrogen atoms reaches first excitation state.

[Ans: 20.4 eV]

2. In the Bohr atom model, the frequency of transitions is given by the following expression

$v = Rc\left(\frac{1}{n^2} - \frac{1}{m^2}\right)$, where $n < m$,

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